Math 372: Homework 03

Due Wednesday, Feb 4

Problem 1. The year is 1532, and 31-year old Girolamo Cardano has been appointed mathematics chair at the University of Milan. In celebration, Cardano decides to host a dinner party. In his cellar he finds 30 bottles of wine: 8 bottles of zinfandel, 10 bottles of merlot, and 12 bottles of cabernet, all from different wineries. (Mr. Cardano only drinks red wine).

(a) He wants to serve 3 bottles of zinfandel and the serving order is important, how many ways are there to do this?

(b) Suppose 6 bottles are randomly selected. How many ways are there to do this?

(c) If 6 bottles are randomly selected, how many ways are there to obtain two bottles of each variety?

(d) What's the probability of that happening?

(e) If 6 bottles are randomly selected, what's the probability that they are all of the same variety (i.e., all zinfandel, all merlot, or all cabernet)?

Problem 2 (Exercise 2.3.34)**.** Computer keyboard failures can be attributed to electrical defects or mechanical defects. A repair facility currently has 25 failed keyboards, 6 of which have electrical defects and 19 of which have mechanical defects.

(a) How many ways are there to randomly select 5 of these keyboards for a thorough inspection (without regard to order)?

(b) In how many ways can a sample of 5 keyboards be selected so that exactly 2 have an electrical defect?

(c) If a sample of 5 keyboards is randomly selected, what is the probability that at least 4 of these will have a mechanical defect?

Problem 3. We return to the dice-rolling pirate, Diego. Suppose Diego rolls two dice and then adds the numbers together. The possible outcomes are $2, 3, 4, 5, 6, 7, 8, 9, 10, 11,$ and 12, but recall that these numbers are not all equally likely.

(a) What is the probability that the sum of the two dice is greater than 4?

(b) Given that the sum of the dice is greater than 4, what is the probability that the dice added up to 7?

(c) Diego rolls his dice again. What is the probability that the dice add up to an even number?

(d) Diego rolls his dice once more. What is the probability that the dice add up to an odd number?

(e) Diego rolls his dice another time. Given that the dice add up to an even number, what is the probability that Diego rolled at least one 4?

(f) Diego rolls his dice again, and the two dice sum to 7. What is the probability that he rolled a 3 or a 5?

(g) Diego rolls his dice again. What is the probability that at least one of Diego's dice was a 3?

(h) Given that at least one of Diego's dice was a 3, what is the probability that his dice add up to 8 or 9?

(i) Diego decides to roll his pair of dice another time. What is the probability that they sum to 4 or 7?

(j) As one last experiment Diego decides to keep rolling the two dice until he gets a sum of either 4 or a 7, at which point he will stop and go back to his ship. What's the probability that his last roll was 4?

Problem 4. A vampire goes to the blood bank looking to find some type O+ blood (the most delicious type). He finds 8 unlabeled bags of blood. Only one of the bags is O+, but the vampire doesn't know which one, so he resorts to taste-testing to find the O+ bag.

(a) What is the probability that he must test at least 5 bags to find the desired type?

(b) What's the probability that he must test exactly 5 bags?

Problem 5. A 5-card poker hand is a **full house** it has a 3-of-a-kind and 2-of-a-kind.

(a) How many possible full houses are there?

(b) Suppose you are dealt a random poker hand. What's the probability that you get a full house?